Target information in retained relations

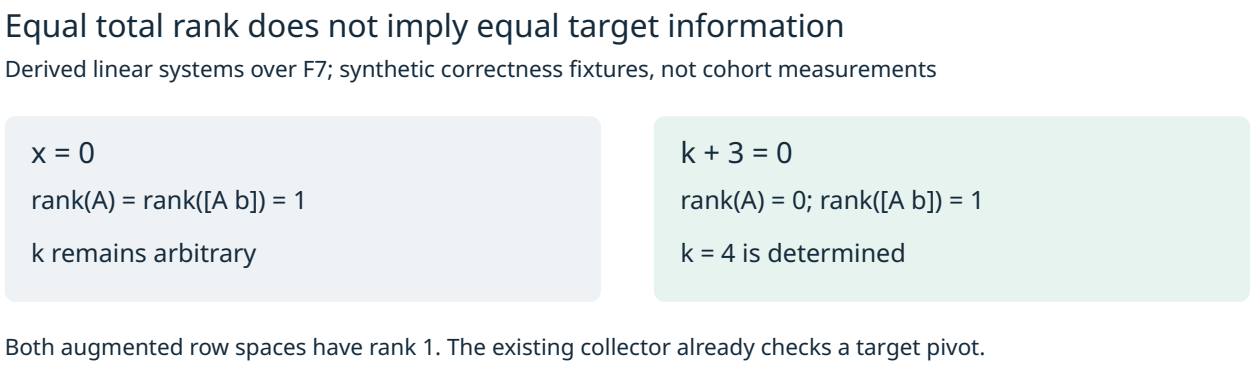
2026-10-30. Derived diagnostic and synthetic correctness controls.

For a consistent system $Ax + bk + c = 0$ over a prime field, the target k is uniquely determined exactly when b is outside the column space of A. Equivalently,

$$\operatorname{rank}[A\ b] - \operatorname{rank} A = 1.$$

Consistency is tested separately by $\operatorname{rank}[A\ b\ c] = \operatorname{rank}[A\ b]$. An inconsistent system is never reported as determining a valid target.

Derivation. Subtract any two solutions. If $b \notin \operatorname{col} A$, a nonzero difference in k would express b as a linear combination of columns of A, a contradiction. Conversely, if $b = Av$, every solution extends to $(x - tv, k + t)$ for every field element t, so k is not unique. Existence is why consistency is checked first.



Source observation. The existing GraphSolver in `native_solver.rs` already checks a target-column pivot after an independent admission (lines 500–516 at the recorded source). This packet does not introduce a new stopping rule. It adds an offline diagnostic to separate nuisance rank from target information.

Implementation and review boundary

The diagnostic reuses the existing prime-field residual basis three times, projecting to nuisance columns, nuisance plus target, and all columns. Rows have a fixed width and composite moduli are rejected by the existing basis constructor. It does not alter collection, elimination order, enrollment, or original results.

Controls. Six native unit tests passed: equal total rank with different information; nuisance-rank growth; inconsistency; row reordering/scaling and empty systems; wrong-domain/shape rejection; and direct enumeration of solution sets for fixed small algebraic fixtures. These are software correctness checks, not scientific samples or a cohort experiment. Native fixture output retains all five example matrices.

Cost boundary. Three basis computations incur additional work and mutate the kernel's thread-local operation counters. The diagnostic must run outside a measured producer. No time, memory advantage or end-to-end ratio is claimed. Replaying a fixed relation stream cannot establish a changed collector policy's future collection cost.

Remaining evidence. Independent source review is pending. This packet does not reconstruct per-relation information histories from aggregate historical rank counters. A subsequent policy comparison needs an actual retained relation stream, or prospectively reviewed instrumentation and a separate frozen evaluation. Existing 1,260-target obligations remain unchanged.

Artifacts. `result.json` contains formulas, scope and exact fixture results; `information.svg` is editable; source, validation logs and the source manifest accompany the report. No scalar search or new curve experiment was performed.